

Gravity beyond Newton in a boson cloud

Lecture notes: the post-Newtonian (h_{0i} , h_{ij} , ...) corrections to self-gravity-induced level transitions, and how to go beyond the expansion

sg_vs_si project

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Abstract

These notes teach, from scratch, how to compute the first relativistic corrections to the gravitational interaction between the quanta of a boson cloud. The worked example is the $2 \rightarrow 2$ transition $211 \times 211 \rightarrow 322 \times \text{BH}$ and its relatives. We build everything up in steps:

1. linearised gravity;
2. “integrating out” the graviton to obtain an interaction between stress tensors;
3. checking that interaction against the classic post-Newtonian two-body Lagrangian;
4. writing down the cloud’s stress tensor;
5. turning the result into a transition amplitude that the existing Green’s-function code can evaluate.

The last lecture explains how to avoid the post-Newtonian expansion altogether. Numerical results are at the end: the gravitomagnetic (h_{0i}) term is the largest correction, $\simeq -1.1\alpha^2$ in amplitude for the 211/322 channels, and the spatial-metric (h_{ij}) term is $\sim 5\times$ smaller and positive.

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1 Lecture 0 — The question, and what “post-Newtonian” means

1.1 The object of interest

Integrating out the Newtonian gravitational potential turns the cloud’s self-gravity into a two-body (quartic) interaction between its quanta:

$$H_N = \frac{1}{2} \iint d^3x d^3y |\psi(\mathbf{x})|^2 K_N(\mathbf{x} - \mathbf{y}) |\psi(\mathbf{y})|^2, \quad K_N = -\frac{G\mu^2}{|\mathbf{x} - \mathbf{y}|}, \quad (1)$$

where ψ is the non-relativistic (NR) field of the boson (mass μ). This is exactly the structure of an axion self-interaction, which has $K_{SI} = -\lambda \delta^3(\mathbf{x} - \mathbf{y})/8\mu^2$ instead. Expanding $\psi = \sum_N c_N \psi_N$ in cloud eigenstates, (1) contains terms like $\bar{c}_4 \bar{c}_3 c_1 c_1$. These describe two quanta in state 1 scattering into state 3 plus a state 4 that falls into the black hole (BH) or escapes. The SG-versus-SI comparison computed the rate of such processes with K_N . **The question of these notes: what does gravity look like beyond K_N , and how large are the corrections?**

1.2 What “post-Newtonian” means

Newtonian gravity is the leading term of general relativity in two small, linked quantities:

$$\epsilon_v \equiv \frac{v^2}{c^2}, \quad \epsilon_\Phi \equiv \frac{|\Phi|}{c^2} = \frac{GM}{rc^2}.$$

For any gravitationally bound system the virial theorem ties them together, $v^2 \sim GM/r$, so they are the same order. The post-Newtonian (PN) expansion is the expansion of GR in this one parameter: “1PN” means corrections of relative order v^2/c^2 .

How big is 1PN for the cloud?

A quantum in $|n\ell m\rangle$ has typical speed $v \sim \alpha/n$ (hydrogen: $v = \alpha_{EM}/n$) and sits at $r \sim n^2/(\mu\alpha) = n^2 GM/\alpha^2$, so

$$\frac{v^2}{c^2} \sim \frac{GM}{r} \sim \frac{\alpha^2}{n^2}.$$

At $\alpha = 0.3$ and $n = 2, 3$ this is 1–2%, multiplied by $O(1)$ – $O(10)$ coefficients. We should therefore expect corrections of a few to $\sim 20\%$, smaller for higher- n states. (From here on $c = \hbar = 1$.)

1.3 The analogy to keep in mind: the Breit interaction

In atomic physics, the Coulomb interaction e^2/r between two electrons is the “Newtonian” term. Its first relativistic correction is the *Breit interaction*, which has two pieces:

- a *magnetic* (current–current) interaction, because moving charges are currents;
- a *retardation* correction, because the photon takes time to travel.

Gravity is the same story with two twists. The “charge” is energy–momentum, a tensor rather than a vector, so there are more pieces. And the field gravitates itself, so there are nonlinear pieces. The gravitational analogue of magnetism is *gravitomagnetism*, carried by the h_{0i} components of the metric.

2 Lecture 1 — Linearised gravity in one sitting

2.1 The metric perturbation and how matter couples to it

We write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $\eta = \text{diag}(-1, +1, +1, +1)$ and $|h| \ll 1$. Matter is described by an action $S_m[g]$, and its stress–energy tensor is *defined* by how S_m responds to the metric:

$$T^{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g_{\mu\nu}}.$$

Expanding $S_m[\eta + h]$ to first order in h therefore gives the coupling

$$S_m[\eta + h] = S_m[\eta] + \frac{1}{2} \int d^4x h_{\mu\nu} T^{\mu\nu} + O(h^2). \quad (2)$$

(The $\frac{1}{2}$ comes straight from the 2 in the definition.)

2.2 The field equation

The Einstein–Hilbert action expanded to second order in h is the Fierz–Pauli action for a massless spin-2 field. It has a gauge freedom, $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$: infinitesimal coordinate changes, the analogue of $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$. Two definitions make life simple:

- the **trace-reversed** field $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, with $h = \eta^{\mu\nu}h_{\mu\nu}$ (note $\bar{\bar{h}} = -h$);
- the **harmonic** (de Donder, Lorenz) gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$.

In this gauge the linearised Einstein equations become ten decoupled wave equations,

$$\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}, \quad \square = -\partial_t^2 + \nabla^2, \quad (3)$$

which is Maxwell’s $\square A_\mu = -J_\mu$ with one extra index.

2.3 Static solution, and undoing the trace reversal

If the source changes slowly, drop ∂_t^2 . Since $\nabla^2(1/|\mathbf{x} - \mathbf{y}|) = -4\pi\delta^3(\mathbf{x} - \mathbf{y})$, the solution is

$$\bar{h}_{\mu\nu}(\mathbf{x}) = 4G \int d^3y \frac{T_{\mu\nu}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}. \quad (4)$$

To get back $h_{\mu\nu}$, invert the trace reversal, $h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}$, with $\bar{h} = -\bar{h}_{00} + \bar{h}_{kk}$:

$$h_{00} = \frac{1}{2}(\bar{h}_{00} + \bar{h}_{kk}), \quad h_{0i} = \bar{h}_{0i}, \quad h_{ij} = \bar{h}_{ij} + \frac{1}{2}\delta_{ij}(\bar{h}_{00} - \bar{h}_{kk}). \quad (5)$$

2.4 The Newtonian limit, and three lessons

For slow, non-relativistic matter, $T_{00} = \rho$ is the only large component. With the Newtonian potential $\Phi(\mathbf{x}) = -G \int \rho/|\mathbf{x} - \mathbf{y}|$, Eqs. (4)–(5) give $\bar{h}_{00} = -4\Phi$ and

$$h_{00} = -2\Phi, \quad h_{ij} = -2\Phi\delta_{ij}, \quad ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)d\mathbf{x}^2. \quad (6)$$

Lesson 1: even Newtonian gravity curves space

h_{ij} is as large as h_{00} ! Yet planets don't feel it at leading order. A particle's action is $-m \int d\tau$, with

$$d\tau = dt \sqrt{(1 + 2\Phi) - (1 - 2\Phi)v^2} \simeq dt \left(1 + \Phi - \frac{1}{2}v^2 + \dots\right).$$

So h_{ij} always comes multiplied by $v^i v^j$, and its effects are suppressed by v^2 . **The h_{ij} correction we are after is the spatial curvature that Newtonian gravity already produces, felt through the cloud's internal motion.**

Lesson 2: moving mass makes gravitomagnetism

$\bar{h}_{0i}$ is sourced by $T_{0i} = -\rho v_i$, the momentum density:

$$h_{0i} = -4G \int \frac{\rho v_i}{|\mathbf{x} - \mathbf{y}|}.$$

This is a vector potential, sourced by mass currents exactly as $\mathbf{A}$ is sourced by electric currents. It couples to a moving particle as $h_{0i} T^{0i} \propto h_{0i} v^i$, so h_{0i} , which is already $O(v)$, enters the dynamics at $O(v^2)$, the same order as h_{ij} . (The “4” instead of EM's “1” is a spin-2 factor.)

Lesson 3: retardation

The exact solution of (3) is retarded: $\bar{h}(t, \mathbf{x}) = 4G \int T(t - |\mathbf{x} - \mathbf{y}|, \mathbf{y}) / |\mathbf{x} - \mathbf{y}|$. Taylor-expanding the time argument,

$$\frac{T(t - r)}{r} = \frac{T(t)}{r} - \partial_t T + \frac{r}{2} \partial_t^2 T - \dots$$

The middle term integrates to a spatial constant. The last is suppressed by $(\partial_t r)^2 \sim v^2$, which is again 1PN. For a source oscillating as $e^{-ik_0 t}$ the replacement is $1/r \rightarrow 1/r - \frac{1}{2} k_0^2 r$, the expansion of $e^{ik_0 r}/r$, or of $1/(\mathbf{k}^2 - k_0^2)$ in momentum space.

3 Lecture 2 — Integrating out the graviton

3.1 A toy model first

Take a scalar field coupled to a source, $L = -\frac{1}{2}(\partial\phi)^2 + J\phi$. The static solution is $\phi = \int J/(4\pi r)$. Put it back into L , integrating $-\frac{1}{2}(\nabla\phi)^2 = \frac{1}{2}\phi \nabla^2 \phi = -\frac{1}{2}J\phi$ by parts:

$$L_{\text{on shell}} = -\frac{1}{2}J\phi + J\phi = \frac{1}{2} \int J\phi = \frac{1}{2} \iint \frac{J(\mathbf{x})J(\mathbf{y})}{4\pi|\mathbf{x} - \mathbf{y}|}.$$

Rule

For any field with a quadratic action and a linear coupling $\int(\text{field}) \times (\text{source})$, the on-shell (“integrated-out”) action is **one half of the coupling term** evaluated on the solution. It is a source–source interaction mediated by the field's Green's function. A positive L means a negative energy, i.e. attraction.

3.2 Apply it to gravity

The coupling is $\frac{1}{2} \int h_{\mu\nu} T^{\mu\nu}$, so $S_{\text{eff}} = \frac{1}{2} \times \frac{1}{2} \int h_{\mu\nu} [T] T^{\mu\nu}$. Using (4) and the trace reversal, $h_{\mu\nu} = 4G \int [T_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} T] / r$. Therefore

$$L_{\text{eff}} = G \iint \frac{T_{\mu\nu}(\mathbf{x}) T^{\mu\nu}(\mathbf{y}) - \frac{1}{2} T(\mathbf{x}) T(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} \quad (+ \text{retardation}). \quad (7)$$

The combination $T_{\mu\nu} T'^{\mu\nu} - \frac{1}{2} T T'$ is the hallmark of spin-2 exchange (spin-1 would give $J_\mu J^\mu$).

3.3 Unpacking the indices (the step to do slowly)

With $\eta = \text{diag}(-1, 1, 1, 1)$, raising a time index flips the sign, so $T^{00} = T_{00}$, $T^{0i} = -T_{0i}$, $T^{ij} = T_{ij}$, and the trace is $T = -T_{00} + T_{kk}$. Then

$$\begin{aligned} T_{\mu\nu} T'^{\mu\nu} &= T_{00} T'_{00} - 2T_{0i} T'_{0i} + T_{ij} T'_{ij}, \\ \frac{1}{2} T T' &= \frac{1}{2} (T_{00} T'_{00} - T_{00} T'_{kk} - T_{kk} T'_{00} + T_{kk} T'_{kk}), \end{aligned}$$

and the difference is

$$\frac{1}{2} T_{00} T'_{00} + \frac{1}{2} (T_{00} T'_{kk} + T_{kk} T'_{00}) - 2T_{0i} T'_{0i} + [T_{ij} T'_{ij} - \frac{1}{2} T_{kk} T'_{kk}].$$

Power counting. For matter moving at speed v : $T_{00} \sim \rho$, $T_{0i} \sim \rho v$, $T_{ij} \sim \rho v^2$. The square bracket is $O(\rho^2 v^4)$, which is 2PN, so we drop it:

$$L_{\text{eff}} = G \iint \frac{1}{r} \left[\underbrace{\frac{1}{2} T_{00} T'_{00}}_{(a)} + \underbrace{\frac{1}{2} (T_{00} T'_{kk} + T_{kk} T'_{00})}_{(b)} - \underbrace{2 T_{0i} T'_{0i}}_{(c)} \right] + (d) \text{retardation} + O(v^4). \quad (8)$$

Term (a) with $T_{00} = \rho$ is Newton, $L = \frac{G}{2} \iint \rho \rho' / r = -V$. At 1PN it also carries the $O(v^2)$ part of T_{00} itself.

3.4 Which metric component is responsible for which term?

Go back to $S_{\text{eff}} = \frac{1}{4} \int [h_{00} T^{00} + 2h_{0i} T^{0i} + h_{ij} T^{ij}]$ and use (5):

metric piece	sourced by	gives
$h_{00} \supset \frac{1}{2} \bar{h}_{00}$	T_{00}	(a): Newton (+ v^2 part of T_{00})
$h_{00} \supset \frac{1}{2} \bar{h}_{kk}$	T_{kk} (stress, “pressure”)	half of (b)
$h_{ij} \supset \frac{1}{2} \delta_{ij} \bar{h}_{00}$	T_{00} , coupling to T^{ij}	other half of (b)
$h_{0i} = \bar{h}_{0i}$	T_{0i} (mass current)	(c): gravitomagnetism
time dependence of $\bar{h}_{00}$	$\partial_t^2 T_{00}$	(d): retardation

The two halves of (b) are equal under $\mathbf{x} \leftrightarrow \mathbf{y}$. Physically:

- h_{ij} : the space curvature of Newtonian gravity acting on the cloud’s internal stresses (Lesson 1).
- $h_{00}[T_{kk}]$: pressure gravitates, $\rho \rightarrow \rho + 3p$ in the active mass.

Note that Bošković–Savić (v2, Eqs. 27–29) list only the leading sources, $\nabla^2 h_{00} = -8\pi T^{00}$, $\nabla^2 h_{0i} = 16\pi T^{0i}$, $\nabla^2 h_{ij} = -8\pi \delta_{ij} T^{00}$. At 1PN, h_{00} also gets the T_{kk} source.

3.5 Checking everything on two point masses

Before using (8) on a cloud, let's make sure it reproduces the textbook answer. For point particles,

$$T^{00} = m\gamma\delta^3 \simeq m(1 + \tfrac{1}{2}v^2)\delta^3, \quad T^{0i} = mv^i\delta^3, \quad T^{ij} = mv^iv^j\delta^3.$$

The double integral visits the pair twice, as (1, 2) and (2, 1). Term by term:

$$\begin{aligned} \text{(a)} : & \frac{G}{2} \cdot 2 \cdot \frac{m_1 m_2}{r} (1 + \tfrac{1}{2}v_1^2)(1 + \tfrac{1}{2}v_2^2) \simeq \frac{Gm_1 m_2}{r} \left[1 + \tfrac{1}{2}(v_1^2 + v_2^2)\right], \\ \text{(b)} : & \frac{Gm_1 m_2}{r} (v_1^2 + v_2^2), \\ \text{(c)} : & -2G \cdot 2 \cdot \frac{(m_1 \mathbf{v}_1) \cdot (m_2 \mathbf{v}_2)}{r} = -\frac{4Gm_1 m_2}{r} \mathbf{v}_1 \cdot \mathbf{v}_2, \\ \text{(d)} : & \frac{Gm_1 m_2}{2r} [\mathbf{v}_1 \cdot \mathbf{v}_2 - (\mathbf{n} \cdot \mathbf{v}_1)(\mathbf{n} \cdot \mathbf{v}_2)] \quad (\text{after removing a total time derivative}). \end{aligned}$$

Sum = Einstein–Infeld–Hoffmann

Adding up:

$$L_{2\text{-body}} = \frac{Gm_1 m_2}{r} \left[1 + \tfrac{3}{2}(v_1^2 + v_2^2) - \tfrac{7}{2} \mathbf{v}_1 \cdot \mathbf{v}_2 - \tfrac{1}{2}(\mathbf{n} \cdot \mathbf{v}_1)(\mathbf{n} \cdot \mathbf{v}_2)\right],$$

which is exactly the velocity-dependent part of the EIH Lagrangian (Landau–Lifshitz §106) in harmonic coordinates. So the four mechanisms (a)–(d), with the coefficients above, are right. Note how the famous $-\frac{7}{2}$ splits: -4 from gravitomagnetism and $+\frac{1}{2}$ from retardation.

3.6 The piece linear-gravity misses: G^2 terms

EIH also contains terms quadratic in G , which do *not* come from (7):

$$L_{G^2} = -\frac{G^2}{2} \sum_a \sum_{b \neq a} \sum_{c \neq a} \frac{m_a m_b m_c}{r_{ab} r_{ac}}.$$

They arise from nonlinearities. The particle action $-m \int d\tau$ contains $\hbar^2$ terms, and gravity's cubic self-interaction lets the field of one body scatter off the field of another. For a cloud around a hole of mass M , keep the terms with two cloud quanta and the BH. With ρ the cloud's number density this gives

$$L_{G^2} = -\frac{G^2 M \mu^2}{2} \iint |\psi(\mathbf{x})|^2 |\psi(\mathbf{y})|^2 \left[\frac{1}{r_x r_y} + \left(\frac{1}{r_x} + \frac{1}{r_y} \right) \frac{1}{|\mathbf{x} - \mathbf{y}|} \right]. \quad (9)$$

Why this is *not* higher order

Relative to Newton, (9) is suppressed by GM/r , the BH's potential at the cloud. By the virial theorem this is the same size as v^2 , so it is 1PN. Physically, the graviton exchanged between two cloud quanta propagates through the BH's gravitational field.

Gauge dependence

Each of (a)–(d) and L_{G^2} depends on the harmonic-gauge choice; only their total, combined with consistent coordinates for the wavefunctions, is physical. This is just like the Breit interaction:

in Coulomb vs Feynman gauge the split between “magnetic” and “retardation” differs, but the on-shell sum agrees. Individual pieces are still useful *indicators of size*.

4 Lecture 3 — The stress tensor of the cloud

4.1 From the relativistic field to ψ

The axion is a real scalar with $S = \int \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \mu^2 \phi^2 \right]$, so

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - g_{\mu\nu} \left[\frac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi + \frac{1}{2} \mu^2 \phi^2 \right].$$

Bound modes oscillate at $\omega \simeq \mu$. We factor that out:

$$\phi = \frac{1}{\sqrt{2\mu}} (\psi e^{-i\mu t} + \bar{\psi} e^{+i\mu t}), \quad |\dot{\psi}| \sim \mu \alpha^2 |\psi|, \quad |\nabla \psi| \sim \mu \alpha |\psi|.$$

The $1/\sqrt{2\mu}$ makes ψ obey the Schrödinger equation with $\int |\psi|^2 = \text{number of quanta}$.

Throwing away the $e^{\pm 2i\mu t}$ terms

Products like $\phi\phi$ contain pieces $\propto e^0$ and $\propto e^{\pm 2i\mu t}$. The latter would source gravitons of frequency 2μ . Those are real, radiated gravitons: the annihilation process $aa \rightarrow g$, a different physical effect. For the potential exchange between bound quanta we keep only the slowly varying parts.

4.2 The three components

Write $\langle \cdot \rangle$ for the slowly varying part. With $\dot{\phi} = \frac{1}{\sqrt{2\mu}} [(-i\mu\psi + \dot{\psi})e^{-i\mu t} + \text{c.c.}]$:

$$\langle \dot{\phi}^2 \rangle = \frac{1}{\mu} [\mu^2 |\psi|^2 + i\mu(\bar{\psi}\dot{\psi} - \psi\dot{\bar{\psi}}) + |\dot{\psi}|^2], \quad \langle (\nabla \phi)^2 \rangle = \frac{1}{\mu} |\nabla \psi|^2, \quad \langle \mu^2 \phi^2 \rangle = \mu |\psi|^2.$$

This gives:

$$T_{00} = \mu |\psi|^2 + \frac{i}{2} (\bar{\psi}\dot{\psi} - \psi\dot{\bar{\psi}}) + \frac{1}{2\mu} |\nabla \psi|^2, \quad (10)$$

$$T_{0i} = \frac{i}{2} (\bar{\psi}\partial_i \psi - \psi\partial_i \bar{\psi}) \equiv -j_i, \quad \mathbf{j} = -\frac{i}{2} (\bar{\psi}\nabla \psi - \psi\nabla \bar{\psi}), \quad (11)$$

$$T_{kk} = \frac{1}{\mu} |\nabla \psi|^2 - \frac{3}{4\mu} \nabla^2 |\psi|^2. \quad (12)$$

Physical meaning

- For a plane wave $\psi = \sqrt{n} e^{i\mathbf{k}\cdot\mathbf{x}}$: $\mathbf{j} = n\mathbf{k} = \mu n\mathbf{v}$, the **mass current**; and $T_{kk} = nk^2/\mu = \mu n v^2 = \rho v^2$, as for particles.
- For a bound state, (12) integrates to $\int T_{kk} = \frac{1}{\mu} \int |\nabla \psi|^2 = 2K$, twice the kinetic energy, exactly $\sum mv^2$.

One subtle point about T_{kk}

A flat-space calculation gives $T_{kk} = -\frac{1}{2\mu} |\nabla \psi|^2 + 3\varepsilon |\psi|^2$, where ε is the mode’s energy, kinetic plus *potential* ($V = -GM\mu/r$). That contains $3V|\psi|^2$ and gives $\int T_{kk} = 2K + 3W$, the wrong

answer for a bound state. The cure is to keep the BH's metric factors g_{00}, g_{ij} in $T_{\mu\nu}$. At $O(\Phi_{\text{BH}})$ they remove the V term and leave the purely kinetic (12). (Potential-energy pieces of this kind belong with the G^2 terms; we must not count them twice.)

The same remark applies to the $O(v^2)$ part of T_{00} , whose split between “kinetic” and “potential” and between T_{00} and the normalisation of ψ is convention dependent. Below we use the particle-like choice $T_{00}^{(1)} = \frac{1}{2\mu} \nabla \bar{\psi} \cdot \nabla \psi$, the kinetic energy density.

4.3 Transition densities

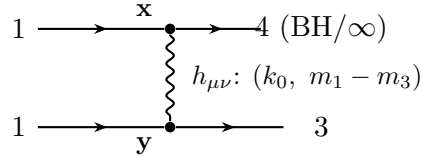
In a $2 \rightarrow 2$ process the graviton is emitted when one quantum makes a transition $A \rightarrow B$. Since $T_{\mu\nu}$ is bilinear in ψ , the relevant objects are the cross terms between modes:

$$\rho^{BA} = \bar{\psi}_B \psi_A, \quad \mathbf{j}^{BA} = -\frac{i}{2} (\bar{\psi}_B \nabla \psi_A - \psi_A \nabla \bar{\psi}_B), \quad t^{BA} \equiv \mu T_{kk}^{BA} = \nabla \bar{\psi}_B \cdot \nabla \psi_A - \frac{3}{4} \nabla^2 (\bar{\psi}_B \psi_A).$$

These oscillate as $e^{-i(\omega_A - \omega_B)t}$. So the graviton carries energy $k_0 = \omega_A - \omega_B$ (a level spacing, tiny) and azimuthal number $m_A - m_B$.

5 Lecture 4 — From L_{eff} to the transition amplitude

5.1 Reading off the amplitude



Insert $\psi = \sum c_N \psi_N$ into $H_{\text{int}} = -L_{\text{eff}}$ and keep the terms $\propto \bar{c}_4 \bar{c}_3 c_1 c_1$. They come from putting the bilinear (41) at one end of the graviton and (31) at the other, in both orders. Every term below has the same overall combinatorial factor as the Newtonian and SI vertices, so that factor cancels in all ratios:

$$A_{\text{N}} = -G\mu^2 \iint \frac{\rho^{41}(\mathbf{x}) \rho^{31}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}, \quad (13)$$

$$A_{h_{0i}} = +4G \iint \frac{\mathbf{j}^{41}(\mathbf{x}) \cdot \mathbf{j}^{31}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}, \quad (14)$$

$$A_{h_{ij}} = A_{h_{00}[T_{kk}]} = -\frac{G}{2} \iint \frac{\rho^{41} t^{31'} + t^{41} \rho^{31'}}{|\mathbf{x} - \mathbf{y}|}, \quad (15)$$

$$A_{T_{00}} = -\frac{G}{2} \iint \frac{\rho^{41} (\nabla \bar{\psi}_3 \cdot \nabla \psi_1)' + (\nabla \bar{\psi}_4 \cdot \nabla \psi_1) \rho^{31'}}{|\mathbf{x} - \mathbf{y}|}, \quad (16)$$

$$A_{\text{ret}} = +\frac{G\mu^2 k_0^2}{2} \iint \rho^{41} \rho^{31'} |\mathbf{x} - \mathbf{y}|, \quad k_0 = \omega_1 - \omega_3, \quad (17)$$

$$A_{G^2} = +G^2 M \mu^2 \iint \rho^{41} \rho^{31'} \left[\frac{1}{r_x r_y} + \left(\frac{1}{r_x} + \frac{1}{r_y} \right) \frac{1}{|\mathbf{x} - \mathbf{y}|} \right]. \quad (18)$$

Signs and factors in three lines

- **Newton:** $-L \supset -\frac{G}{2} \iint \mu^2 |\psi|^2 |\psi'|^2 / r$, and the two orderings give the factor 2, hence $-G\mu^2$. This reproduces K_N .
- **Gravitomagnetism:** $-L \supset +2G \iint T_{0i} T'_{0i} / r = 2G \iint \mathbf{j} \cdot \mathbf{j}' / r$, and the two orderings give $+4G$.
- **Retardation:** $1/r \rightarrow -\frac{1}{2} k_0^2 r$ inside A_N . The energy through the graviton is the same, $\pm k_0$, in both orderings, because $2\omega_1 = \omega_3 + \omega_4$.

We report every correction as $\delta_X \equiv A_X/A_N$, so $\Gamma_{\text{SG}} \rightarrow \Gamma_{\text{SG}} |1 + \sum_X \delta_X|^2$.

5.2 Why we must integrate by parts

Mode 4 is not a normalisable bound state. It is defined through a Green's function: the existing code (`gf_radial`) solves the Klein–Gordon equation on Kerr with a *source* $S(\mathbf{x})$ and BH-ingoing or outgoing boundary conditions, and reads off the flux. So we need each amplitude in the form

$$A_X = \int d^3x \bar{\psi}_4(\mathbf{x}) S_X(\mathbf{x}),$$

with no derivatives acting on ψ_4 . Wherever $\nabla \bar{\psi}_4$ appears, we move the derivative onto the rest by parts. Here are the two worked cases.

Gravitomagnetic term. Define the gravitomagnetic vector potential of the transition current, $\mathbf{A}(\mathbf{x}) = \int \mathbf{j}^{31}(\mathbf{y})/|\mathbf{x}-\mathbf{y}|$, so $A_{h_{0i}} = 4G \int \mathbf{j}^{41} \cdot \mathbf{A}$. With $\mathbf{j}^{41} = -\frac{i}{2}(\bar{\psi}_4 \nabla \psi_1 - \psi_1 \nabla \bar{\psi}_4)$ and $\int \psi_1 \mathbf{A} \cdot \nabla \bar{\psi}_4 = -\int \bar{\psi}_4 \nabla \cdot (\psi_1 \mathbf{A})$:

$$\int \mathbf{j}^{41} \cdot \mathbf{A} = -\frac{i}{2} \int \bar{\psi}_4 [2\mathbf{A} \cdot \nabla \psi_1 + \psi_1 \nabla \cdot \mathbf{A}].$$

The divergence is fixed by **particle-number conservation**. The transition current obeys $\partial_t(\mu \rho^{31}) + \nabla \cdot \mathbf{j}^{31} = 0$, and $\rho^{31} \propto e^{-ik_0 t}$, so $\nabla \cdot \mathbf{j}^{31} = i\mu k_0 \rho^{31}$ and hence $\nabla \cdot \mathbf{A} = i\mu k_0 \Phi$ with $\Phi \equiv \int \rho^{31}/|\mathbf{x}-\mathbf{y}|$. Altogether

$$S_{h_{0i}} = -4iG \mathbf{A} \cdot \nabla \psi_1 + 2G\mu k_0 \psi_1 \Phi. \quad (19)$$

Stress term, and where a contact term comes from. Two identities do all the work. The first is $\int f \nabla \bar{\psi}_4 \cdot \nabla \psi_1 = -\int \bar{\psi}_4 \nabla \cdot (f \nabla \psi_1)$. The second is the Poisson equation in the form $\int \nabla^2 \rho(\mathbf{y})/|\mathbf{x}-\mathbf{y}| = -4\pi \rho(\mathbf{x})$. The $\nabla^2(\bar{\psi}\psi)$ piece of t^{BA} is a total derivative, but integrated against $1/|\mathbf{x}-\mathbf{y}|$ it becomes a *local* term. Doing both halves of (15):

$$2S_{h_{ij}} = G \left[-\psi_1 \int \frac{\nabla \bar{\psi}_3 \cdot \nabla \psi_1}{|\mathbf{x}-\mathbf{y}|} + \nabla \Phi \cdot \nabla \psi_1 + \Phi \nabla^2 \psi_1 - 6\pi \psi_1^2 \bar{\psi}_3 \right]. \quad (20)$$

A curiosity

The last term has exactly the form of the axion's contact interaction, $\psi_1^2 \bar{\psi}_3$. For h_{ij} alone (half of the above) it corresponds to $1/(8f^2) \leftrightarrow 3\pi G$. At 1PN, gravity itself generates a small local quartic coupling of the field.

Newton is simply $S_N = -G\mu^2 \psi_1 \Phi$. The T_{00} , retardation and G^2 sources follow in the same way (code: `all_sources` in `sg_1pn.jl`).

5.3 How the numbers are produced

1. **Potentials by multipoles.** Every potential is a Coulomb-type integral. Expand $1/|\mathbf{x} - \mathbf{y}| = \sum_{LM} \frac{4\pi}{2L+1} Y_{LM}(\hat{\mathbf{x}}) \bar{Y}_{LM}(\hat{\mathbf{y}}) r_{<}^L / r_{>}^{L+1}$, project the density onto Y_{LM} on an (r, θ) grid (Gauss–Legendre in $\cos \theta$), and do two cumulative radial integrals per L . The retardation kernel uses

$$|\mathbf{x} - \mathbf{y}| = \sum_L P_L(\cos \gamma) \left[\frac{r_{<}^{L+2}}{(2L+3)r_{>}^{L+1}} - \frac{r_{<}^L}{(2L-1)r_{>}^{L-1}} \right].$$

2. **Vectors as scalars.** $\mathbf{A}$ is built from the Cartesian spherical components $j_{\pm} = j_x \pm i j_y$ and j_z . Each has a definite azimuthal number ($m \pm 1$ and m), so each is a scalar Coulomb problem.
3. **Projection.** Each source is projected onto mode 4’s harmonic, $\int S \bar{Y}_{\ell_4 m_4} (r^2 + a^2 \cos^2 \theta) d\Omega$. This is the same Kerr “ Σ ” weighting `gf_radial` uses. The resulting radial source is then fed to the unchanged Kerr Green’s function.
4. **Which eigenfunctions.** The corrections are themselves $O(\alpha^2)$, so they are evaluated with hydrogenic eigenfunctions; the error is $O(\alpha^4)$. The Newtonian amplitude they are normalised to uses the same eigenfunctions.

How we know the bookkeeping is right

Replace mode 4 by a *bound* state ($|320\rangle$). Then the original double integrals (14)–(15), with derivatives acting on ψ_4 , can be computed directly. They agree with $\int \bar{\psi}_4 S_X$ to 6×10^{-7} (h_{0i}) and 5×10^{-6} (stress). This tests every integration by parts and the continuity step (`julia sg_1pn.jl` test).

6 Lecture 5 — Estimates first, then numbers

6.1 Back of the envelope

Use the point-particle formulas with $v_1 \sim \alpha/n_1$ and $v_3 \sim \alpha/n_3$ as the speeds “at the two vertices”:

- h_{0i} : $\delta \sim -4 \mathbf{v}_1 \cdot \mathbf{v}_3$. Both transition currents inherit the azimuthal flow of the superradiant state 1, so they are *aligned*. Co-moving mass currents *repel* gravitomagnetically (the $-4 \mathbf{v}_1 \cdot \mathbf{v}_2$ in EIH), so this term weakens the attraction. Estimate: $-4(\alpha/2)(\alpha/3) = -0.67\alpha^2$ for $211 \rightarrow 322$, and $-4(\alpha/3)(\alpha/5) = -0.27\alpha^2$ for $322/544$.
- h_{ij} : $\delta \sim +\frac{1}{2}(v_1^2 + v_3^2)$. Stress adds to the active mass, so this term strengthens the attraction. Estimate: $+0.18\alpha^2$ for $211 \rightarrow 322$, and $+0.08\alpha^2$ for $322/544$.
- **Retardation:** $\delta \sim \frac{1}{2} k_0^2 r^2$, with k_0 a level spacing $\sim \alpha^3/n^3$ in GM units and $r \sim n^2/\alpha^2$. This is small.
- G^2 : $\delta \sim -2GM/r \sim -2\alpha^2/n^2$, negative, of the same order as h_{0i} .

6.2 The numbers

6.3 What we learned

1. h_{0i} is the **biggest 1PN effect**, and it is negative. It reduces Γ_{SG} by $\simeq 20\%$ at $\alpha = 0.3$ for $211 \times 211 \rightarrow 322 \times \text{BH}$ and $322 \times 322 \rightarrow 211 \times \infty$, and by $\simeq 6\%$ for the $322/544$ channels. The higher- n states move more slowly, hence the smaller effect.

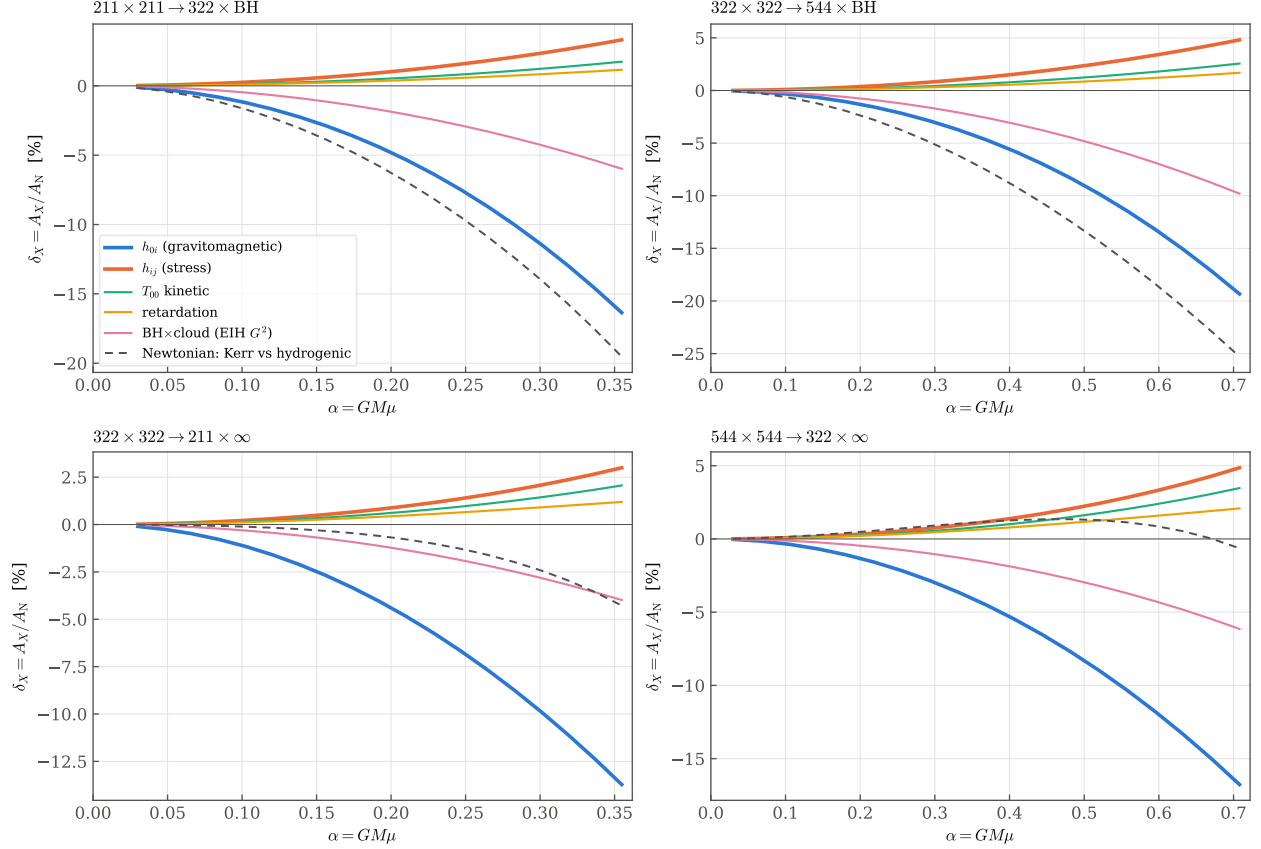


Figure 1: Fractional corrections $\delta_X = A_X/A_N$ to the Newtonian self-gravity amplitude (Kerr Green’s function, $\tilde{a} = 0.95$). Thick lines: h_{0i} and h_{ij} . The h_{ij} line is h_{ij} alone; the stress-sourced h_{00} adds an identical amount. Dashed: change of the *Newtonian* amplitude when Kerr instead of hydrogenic eigenfunctions are used, $\sqrt{C_{\text{Kerr}}/C_{\text{hyd}}} - 1$, a stand-in for the $O(\alpha^2)$ effects on the wavefunction side.

2. h_{ij} is $\sim 5\times$ **smaller and positive**, as estimated.
3. **The G^2 terms are the second largest** and negative. Retardation and T_{00} are at the percent level.
4. **Consequence for SG vs SI.** $\Gamma_{\text{SG}}/\Gamma_{\text{SI}} \propto C$ and the crossover $f_a \propto C^{-1/4}$. The summed correction (-9% in amplitude at $\alpha = 0.3$ for the 211 channel) therefore moves the crossover by only $\lesssim 5\%$, and by $\lesssim 10\%$ over each channel’s full α range.
5. **Accuracy.** The graviton-side corrections are comparable to the wavefunction-side ones (dashed line). Quoting Γ_{SG} to better than $\sim 20\%$ at $\alpha \gtrsim 0.3$ needs a consistent treatment of both, which leads to the last lecture.

channel	h_{0i}	h_{ij}	T_{00}	ret.	G^2 BH $\times$ cloud	sum [†]
$211 \times 211 \rightarrow 322 \times \text{BH}$	-1.14	+0.25	+0.13	+0.10	-0.46	-0.87
$322 \times 322 \rightarrow 544 \times \text{BH}$	-0.32	+0.09	+0.05	+0.03	-0.19	-0.24
$322 \times 322 \rightarrow 211 \times \infty$	-1.10	+0.21	+0.15	+0.12	-0.30	-0.72
$544 \times 544 \rightarrow 322 \times \infty$	-0.33	+0.08	+0.06	+0.05	-0.11	-0.17

Table 1: Small- α coefficients δ_X/α^2 at $\alpha = 0.03$. Compare with the estimates: h_{0i} : -1.1 vs -0.67 and -0.32 vs -0.27; h_{ij} : +0.25 vs +0.18 and +0.09 vs +0.08. [†]Sum = $h_{0i} + 2h_{ij} + T_{00} + \text{ret} + G^2$.

channel	α	$\delta_{h_{0i}}$ [%]	$\delta_{h_{ij}}$ [%]	$\Delta\Gamma_{\text{SG}}(h_{0i})$ [%]	(h_{ij}) [%]	sum [†] [%]
$211 \times 211 \rightarrow 322 \times \text{BH}$	0.30	-11.4	+2.4	-22	+5	-8.9
$322 \times 322 \rightarrow 544 \times \text{BH}$	0.29	-2.9	+0.8	-6	+2	-2.2
$322 \times 322 \rightarrow 211 \times \infty$	0.30	-9.9	+2.1	-19	+4	-6.2
$544 \times 544 \rightarrow 322 \times \infty$	0.29	-2.9	+0.7	-6	+1	-1.4

Table 2: Values at $\alpha \simeq 0.3$; $\Delta\Gamma_{\text{SG}} = |1 + \delta|^2 - 1$.

7 Lecture 6 — Can we avoid the post-Newtonian expansion altogether?

Yes. The PN expansion was only a device to evaluate one object: *the exchange of a single graviton between two transition stress tensors*, in the background spacetime of the black hole. Linear order in h is exact to first order in the cloud mass q_c , which is all we need. What we approximated was the graviton's propagation (flat space plus G^2 corrections) and the stress tensors (expanded in v). Both can be done exactly.

7.1 The exact statement

1. Build the **full transition stress tensor** from the Kerr eigenmodes, with no NR expansion:

$$T_{\mu\nu}^{31} = \partial_{(\mu} \bar{\phi}_3 \partial_{\nu)} \phi_1 - g_{\mu\nu} [\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \bar{\phi}_3 \partial_\beta \phi_1 + \tfrac{1}{2} \mu^2 \bar{\phi}_3 \phi_1],$$

which oscillates at $k_0 = \omega_1 - \omega_3$ with azimuthal number $m_1 - m_3$.

2. **Solve the linearised Einstein equations on Kerr**, $\delta G_{\mu\nu}[h] = 8\pi G T_{\mu\nu}^{31}$, at that single frequency and azimuthal number, with physical boundary conditions (ingoing at the horizon, outgoing at infinity).
3. **Couple to the other vertex.** The perturbed metric changes the Klein–Gordon operator. The source for mode 4 is

$$S_4 = -\delta\Box[h] \phi_1, \quad \delta\Box[h] \phi = -h^{\mu\nu} \nabla_\mu \nabla_\nu \phi - (\nabla_\mu h^{\mu\nu} - \tfrac{1}{2} \nabla^\nu h) \nabla_\nu \phi,$$

and it is fed to the *same* Kerr Green's function as now.

This contains Newton, all PN orders, the G^2 terms (graviton propagation on the curved background) and frame dragging. Being an on-shell amplitude, it is **gauge invariant**, which resolves the ambiguity of the individual PN pieces.

7.2 How to solve step 2 in practice

This is a standard problem of black-hole perturbation theory, the same machinery used for extreme-mass-ratio inspirals and gravitational self-force. Our source is an extended, smooth cloud rather than a point particle, which actually makes it *easier*.

- **Schwarzschild first** ($a = 0$). Decompose h and T in tensor harmonics. The odd and even sectors reduce to the Regge–Wheeler and Zerilli master equations with sources, from which h is reconstructed. Alternatively, solve directly in Lorenz gauge, frequency domain (Barack–Lousto; Akcay). The low multipoles $\ell = 0, 1$ need separate treatment, and they matter: the 211–322 transition density is largely dipolar.
- **Kerr**. Solve the Teukolsky equation for ψ_0 or ψ_4 (spin ± 2) with the source built from $T_{\mu\nu}^{31}$. Then reconstruct h in a radiation gauge (Chrzanowski–Cohen–Kegeles), including the corrector needed inside an extended source (Green–Hollands–Zimmerman) and the mass and angular-momentum completion. Alternatively, use the recently developed separable Lorenz-gauge construction on Kerr (Dolan, Kavanagh, Wardell et al.).
- **Tools**. Black Hole Perturbation Toolkit. `GeneralizedSasakiNakamura.jl`, for spin ± 2 Teukolsky radial functions in Julia, is by the author of `SpinWeightedSpheroidalHarmonics.jl`, which the code already uses. The existing `gw_rates.jl` already solves the *radiative* part of this problem (the flux of ψ_4 to infinity). What is new is the metric *inside* the source region.

7.3 Why it is tractable for us

- **Quasi-static**. The graviton frequency is a level spacing, $k_0 \ll 1/r_{\text{cloud}}$, so we are deep in the near zone and the frequency-domain equations are nearly elliptic.
- **Far from the hole**. The cloud lives at $r \sim n^2 GM/\alpha^2 \gg GM$, so the multipole sums converge fast and the horizon boundary condition matters little.
- **Spin is a small effect**. The Newtonian rate changes by $< 1\%$ between $\tilde{a} = 0.5$ and 0.95 , and spin-dependent gravity starts at 1.5PN . A Schwarzschild calculation of the graviton propagation, combined with the Kerr eigenfunctions and Green’s function already in the code, captures almost everything.

7.4 How we would know it is right

- At small α it must reduce to Newton plus the sum of the 1PN pieces computed here, once the eigenfunctions are expressed in the same (harmonic) coordinates.
- Doing step 2 in two gauges (Regge–Wheeler and Lorenz) must give the same amplitude.

7.5 Other routes, and why we would not take them

- **Flat-space graviton with relativistic stress tensors**. This drops the G^2 terms, which are the same order as everything else, so it is not a consistent improvement.
- **Full numerical relativity of Einstein–Klein–Gordon**. This is used for self-gravitating cloud spectra (e.g. May et al. 2024, which Bošković–Savić compare to). It is exact even beyond linear order in q_c , but extracting $2 \rightarrow 2$ rates over a grid of α is impractically expensive, and linear order in q_c is all we need.

Bottom line

The PN corrections computed here show the size of the effect: $\sim 20\%$ in Γ_{SG} at $\alpha = 0.3$ for the 211 channel, and less for higher n . They are too small to change the SG-vs-SI picture. If percent-level Γ_{SG} is ever needed at large α , the clean route is the exact one-graviton exchange on Kerr: Schwarzschild propagation first, reusing the existing Kerr eigenfunctions and Green's function.

A Sign and index dictionary

- $\eta = \text{diag}(-1, 1, 1, 1)$; $T^{00} = T_{00}$, $T^{0i} = -T_{0i}$, $T^{ij} = T_{ij}$; $T = -T_{00} + T_{kk}$.
- Dust: $T^{00} = \rho$, $T^{0i} = \rho v^i$, so $T_{0i} = -\rho v^i$; $T^{ij} = \rho v^i v^j$.
- $\Phi = -G \int \rho/r$: $h_{00} = -2\Phi$, $h_{ij} = -2\Phi \delta_{ij}$, $g_{00} = -(1 + 2\Phi)$.
- $\nabla^2(1/r) = -4\pi\delta^3$; $\nabla^2 r = 2/r$.
- $L = -V$: a positive term in L_{eff} is attractive.

B Code map

- `sg_1pn.jl`:
 - `hyd_state`: hydrogenic ψ , $\nabla\psi$, $\nabla^2\psi$ on the grid.
 - `coulomb` and `linear_kernel`: the $1/r$ and r kernels.
 - `cart`: spherical $\rightarrow$ Cartesian-spherical components.
 - `all_sources`: S_X for every piece.
 - `point_1pn`: projection and Green's function.
 - `selftest`.
- `sg_vs_si.jl`: `gf_amplitudes`, the Kerr Green's function copied from `gf_radial`.
- `plot_1pn.py`: figure and tables of this note.

C Code fixes made alongside these notes

- `spheroidals` now takes `alph` and uses the massive-field spheroidicity $c = a\sqrt{\omega^2 - \mu^2}$. It previously used $c = a\omega$, the massless convention of `SpinWeightedSpheroidalHarmonics`. Call sites that had passed ω in eV now pass dimensionless energies. Effect on `gf_radial`: $+0.08\%$ to $+2.1\%$.
- The ∞ -channel flux radius in `gf_radial` is now $r(r_*)$, not $r(r(r_*))$ (with the above, $+0.7\%$ for $322 \times 322 \rightarrow 211 \times \infty$ at $\alpha = 0.17$).

Commit `a9bba4b` on `gw-signatures`.